

Radical Halogenation

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1 Foundations

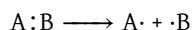
Core Concept 1.1: Radical Halogenation

Radical halogenation is a substitution reaction in which a hydrogen atom of an alkane is replaced by a halogen atom, most commonly chlorine or bromine. Unlike conventional organic reactions that proceed through ionic intermediates, radical halogenation is mediated by **free radicals**—transient, highly reactive species characterized by the presence of an unpaired valence electron.

In synthetic organic chemistry, selective atomic substitution within molecular frameworks is a fundamental objective. However, alkanes present a significant challenge due to their chemical **inertness**; comprising exclusively strong, nonpolar C–C and C–H bonds, alkanes lack reactive functional groups that serve as recognition sites for electrophilic or nucleophilic reagents. Radical halogenation addresses this limitation by converting chemically inert alkanes into functionalized **haloalkanes** (*alkyl halides*), which possess reactive carbon-halogen bonds amenable to subsequent transformations. Consequently, radical halogenation represents a critical entry point into the broader landscape of functional group interconversions.

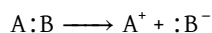
Core Concept 1.2: Homolytic vs. Heterolytic Cleavage

To understand radical formation, we must first examine bond cleavage mechanisms. Consider the **homolytic cleavage** of a generic covalent bond in compound AB.



Homolytic cleavage results in symmetric bond fission, wherein each fragment retains one electron from the original bonding pair, yielding two **radical** species. The enthalpy change (ΔH) for this process corresponds to the bond dissociation energy, typically expressed in units of kcal mol^{-1} and denoted as DH° .

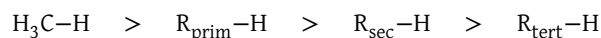
This mechanism is fundamentally distinct from **heterolytic cleavage**, in which the bonding electron pair is retained entirely by one fragment, generating a cation-anion pair:



Importantly, the energy requirement for homolytic cleavage (DH°) typically exceeds that of heterolytic processes in polar solvents, as the latter benefit from solvation stabilization of the resulting ionic species.

Core Concept 1.3: Hydrocarbon C–H Bond Types

Alkane functionalization requires **C–H bond cleavage**, and the bond dissociation energy (DH°) exhibits a systematic decrease across the following homologous series.



The designations R_{prim} , R_{sec} , and R_{tert} refer to primary, secondary, and tertiary carbon centers, which are bonded to one, two, and three additional carbon atoms, respectively. These structural classifications are illustrated in Figure 1.

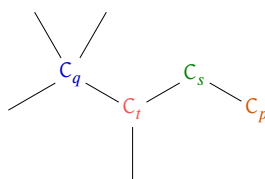


Figure 1: Primary (orange), secondary (green), tertiary (red), and quaternary (blue) carbons

The tertiary $R_{\text{tert}}-\text{H}$ bond exhibits the lowest bond dissociation energy and consequently generates the most thermodynamically stable radical upon cleavage. This stability trend—wherein radical stability increases with increasing alkyl substitution—arises from **hyperconjugation** (Core Concept 1.4).

Core Concept 1.4: Hyperconjugation

Hyperconjugation is a stereoelectronic stabilization mechanism wherein e^- density from a filled σ orbital (typically a $\text{C}-\text{H}$ or $\text{C}-\text{C}$ bond) delocalizes into an adjacent unfilled or partially filled p-orbital. In alkyl radicals, the bonding e^- s of σ bonds on substituent carbon atoms interact with the singly occupied p-orbital (SOMO) of the radical center, resulting in partial e^- delocalization that stabilizes the radical species.

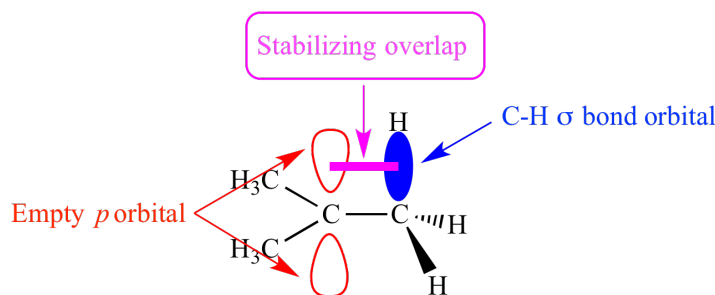


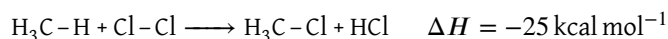
Figure 2: Hyperconjugation in alkane radicals

As illustrated in Figure 2, the magnitude of hyperconjugative stabilization is maximized in tertiary radicals ($\cdot R_{\text{tert}}$) due to the presence of a greater number of adjacent $\text{C}-\text{H}$ or $\text{C}-\text{C}$ σ bonds available for orbital overlap with the radical p-orbital. Each additional alkyl substituent provides additional filled σ orbitals capable of donating e^- density, thereby enhancing radical stability through cumulative hyperconjugative interactions.

2 Mechanism of Radical Halogenation

Core Concept 2.1: Initiation & Propagation

Having established the fundamental principles of alkane radical formation (Section 1), we now examine the mechanistic details of radical halogenation. The overall transformation, illustrated here using chlorine as the halogen, proceeds as follows.



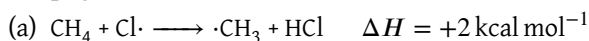
Although the reaction is net exothermic ($\Delta H < 0$), it requires an initial activation energy to proceed. To elucidate the reaction pathway, we can decompose the overall transformation into the following elementary steps.

(1) Initiation



Homolytic cleavage of the $\text{Cl}-\text{Cl}$ bond can be induced through two primary mechanisms. **Thermal energy** elevates molecular vibrational energy to levels sufficient to overcome the bond dissociation energy, resulting in bond scission. Alternatively, **photochemical activation** via UV radiation promotes a bonding e^- to an antibonding molecular orbital ($\sigma \rightarrow \sigma^*$ transition), weakening the $\text{Cl}-\text{Cl}$ bond and facilitating homolytic cleavage.

(2) Propagation



The frontier molecular orbital interactions governing this hydrogen abstraction step are depicted in Figure 3.

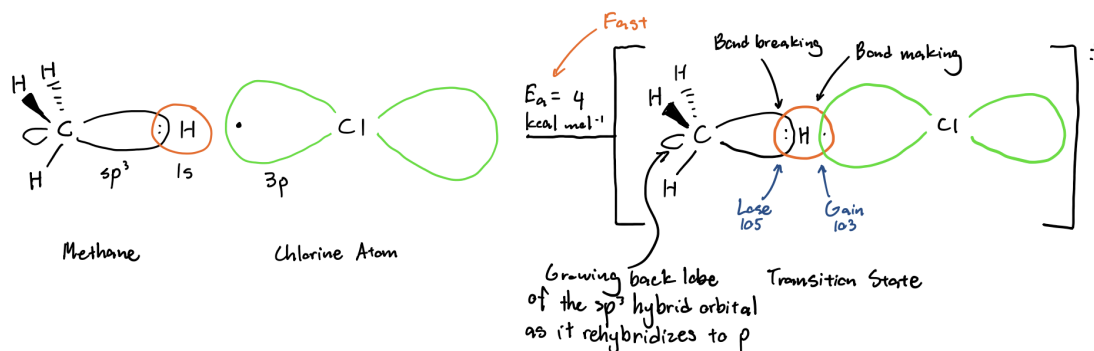


Figure 3: Orbital interaction of reagents in (2a)

This step yields a methyl radical ($\text{H}_3\text{C}\cdot$), characterized by a singly occupied $2p$ orbital and three sp^2 -hybridized C-H bonds adopting trigonal planar geometry, along with HCl. In aqueous or polar environments, HCl, being a strong Brønsted acid, dissociates completely into solvated H^+ and Cl^- ions.

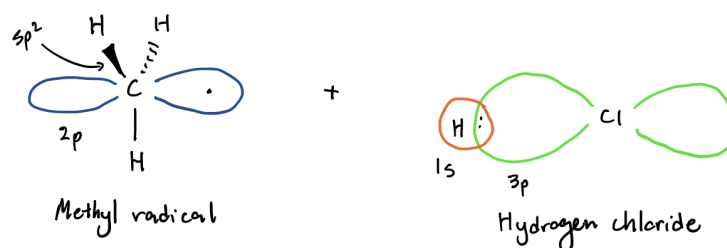
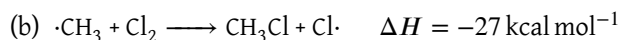


Figure 4: Products of (2a)



Summing the two propagation steps (2a) and (2b) regenerates the original stoichiometric equation.



Notably, the initiation step does not contribute to the net stoichiometry; it functions **solely to generate the initial radical species required to initiate the propagation cycle**. The chain-propagating nature of this mechanism means that a small initial population of $\text{Cl}\cdot$ radicals can facilitate the conversion of substantial quantities of starting materials, as each propagation cycle regenerates the chlorine radical. This autocatalytic regeneration enables each initiating radical to participate in numerous propagation cycles before termination, analogous to a catalytic turnover process.

The PE diagram for the propagation sequence (Figure 5) provides a quantitative representation of the energetic profile throughout the reaction coordinate.

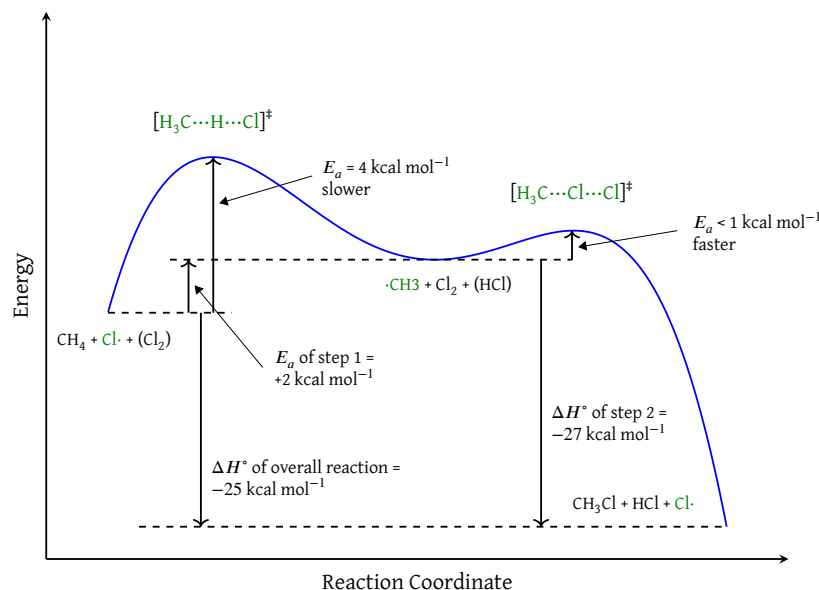
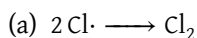


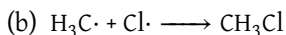
Figure 5: Potential energy diagram of halogenation propagation

Core Concept 2.2: Termination

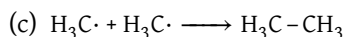
Chain termination occurs through the following radical–radical coupling reactions.



This reaction represents the microscopic reverse of the initiation step, regenerating molecular chlorine through recombination of two chlorine radicals.



While this reaction formally produces the desired chloromethane product, it terminates the chain by consuming a propagating chlorine radical without regeneration, thereby halting further catalytic turnover.



This dimerization reaction couples two methyl radicals to yield ethane, depleting the methyl radical pool required for propagation.

Throughout the reaction, radical intermediates such as $\text{Cl}\cdot$ and $\text{H}_3\text{C}\cdot$ exist at extremely low steady-state concentrations, reacting rapidly with abundant substrate (CH_4 and Cl_2 , respectively) rather than persisting in solution. Consequently, termination reactions—requiring bimolecular radical–radical encounters—are negligible during early and intermediate stages. Termination becomes significant only when substrate depletion reduces propagation rates or when radical concentrations increase sufficiently in the closed system.

3 Halogen Reactivity and Selectivity

Core Concept 3.1: Reactivity of Other Halogens

Having examined chlorination in detail, we now extend our analysis to the full halogen series (*F*, *Br*, and *I*) to assess their respective reactivities.

$DH^\circ (+)$				ΔH of H-X Formation (-)				ΔH of H_3C-X Formation (-)			
F_2	Cl_2	Br_2	I_2	HF	HCl	HBr	HI	CH_3F	CH_3Cl	CH_3Br	CH_3I
38	58	46	36	136	103	87	71	110	85	70	51

Table 1: Bond dissociation and formation enthalpies for halogens and halogen-containing compounds

- **Initiation energetics (left columns):** The bond dissociation energies for X_2 homolysis span a narrow range (36–58 kcal mol⁻¹). Since chlorination proceeds efficiently despite having the highest DH° , initiation is not rate-limiting under typical conditions.
- **Hydrogen halide formation (middle columns):** The exothermicity of H-X bond formation decreases systematically down the halogen group, with ΔH values ranging from -136 kcal mol⁻¹ for HF to -71 kcal mol⁻¹ for HI. HF formation is the most favorable (*most exothermic*) and HI formation is the least favorable (*least exothermic*) among the hydrogen halides.
- **Methyl halide formation (right columns):** An analogous trend is observed for H_3C-X bond formation, with fluorination ($\Delta H = -110$ kcal mol⁻¹) being substantially more exothermic than iodination ($\Delta H = -51$ kcal mol⁻¹).

Based on these thermodynamic data, we can predict that radical halogenation reactivity follows the order: $F > Cl > Br > I$, with fluorination being the most exothermic and iodination the least.

In practice, fluorination is **violently exothermic and uncontrollable**, often resulting in explosion. Conversely, **iodination does not proceed** under standard radical conditions due to its net endothermic character. **Chlorination and bromination occupy a synthetically useful middle ground**: chlorination is more reactive but less selective, while bromination is less reactive but more regioselective (*Core Concept 3.3*).

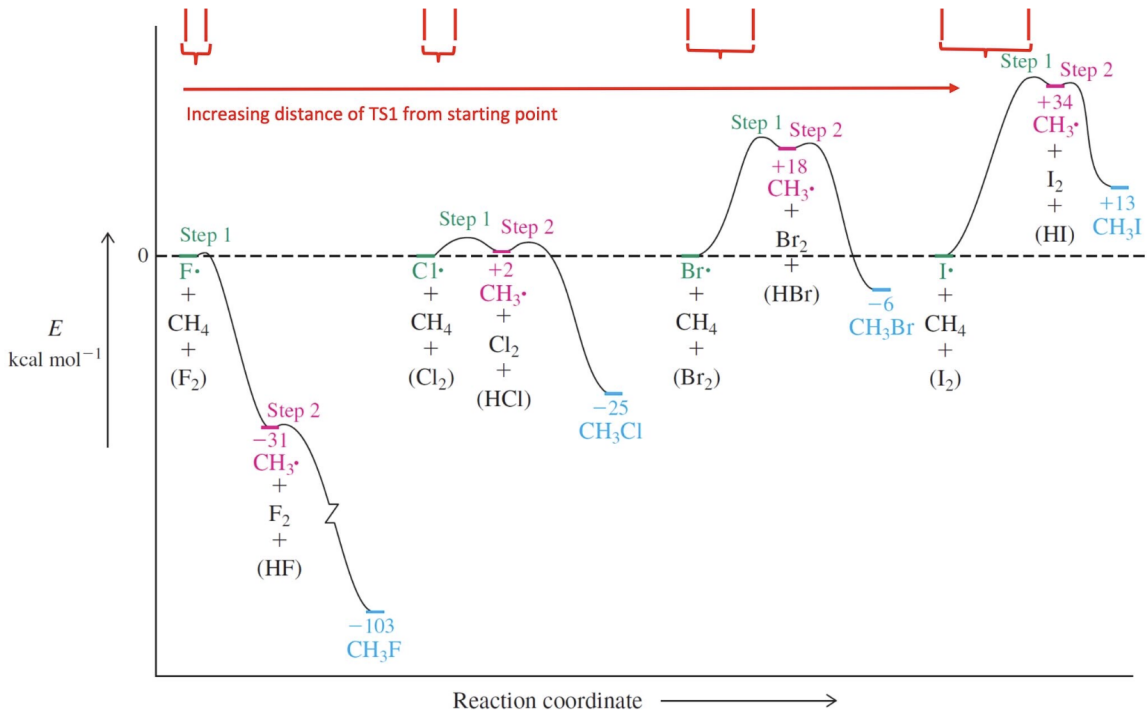


Figure 6: Comparison of Radical Halogenation PE Diagrams

Core Concept 3.2: Hammond Postulate

Inspection of Figure 6 reveals that the position of the first transition state (TS1) along the reaction coordinate shifts systematically as the endothermicity of the first propagation step increases. This observation is rationalized by the **Hammond Postulate**, a fundamental principle that relates transition state structure to reaction thermodynamics. Because transition states are transient, high-energy species that cannot be directly observed, the Hammond Postulate provides a framework for inferring their structures based on energetic considerations.

The Hammond Postulate states that for exothermic elementary steps, the transition state lies energetically and structurally closer to the reactants, resulting in an **early transition state** that resembles the starting materials more than the products.

Conversely, for endothermic elementary steps, the transition state lies closer to the products in both energy and structure, yielding a **late transition state** that more closely resembles the product species.

To illustrate this principle, consider the contrasting cases of fluorination and iodination, shown in magnified detail in Figure 7.

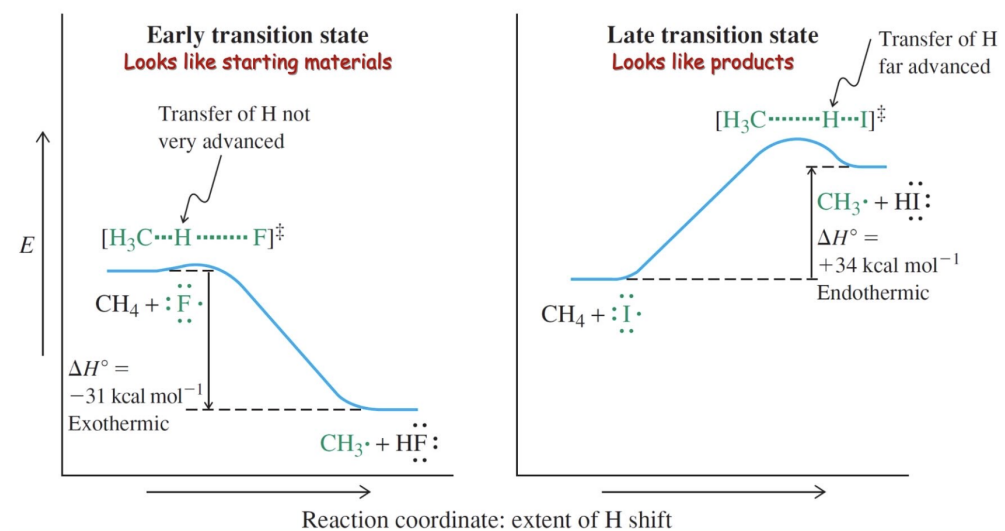


Figure 7: Zoomed Step 1 PE Diagrams for F and I

In the fluorination TS, the transferring hydrogen atom remains significantly closer to the methyl carbon than to the fluorine atom, reflecting the early, reactant-like character of this highly exothermic step. In contrast, the iodination TS exhibits the hydrogen atom positioned much closer to iodine, indicative of a late, product-like TS consistent with the endothermic nature of this step.

This structural distinction correlates directly with E_a differences. Fluorination exhibits a low E_a , requiring minimal C–H bond elongation to reach its TS; consequently, the C–H bond is only slightly weakened when the TS is achieved, and the hydrogen has not yet substantially transferred to fluorine. Iodination, conversely, requires a substantially higher activation energy, necessitating extensive C–H bond elongation and near-complete hydrogen transfer to iodine before the TS is reached. This reflects the weaker H–I bond strength relative to H–F, which requires greater geometric reorganization to compensate for reduced thermodynamic driving force.

Bromination, being mildly endothermic, similarly exhibits a late transition state, though less product-like than that of iodination.

Core Concept 3.3: Regioselectivity

The Hammond Postulate (*Core Concept 3.2*) also provides a mechanistic basis for understanding the differing **regioselectivities** exhibited by various halogens. To illustrate this, consider the competing pathways leading to primary versus tertiary isobutyl radicals:

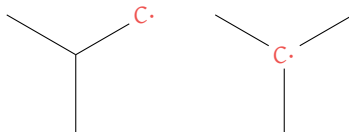


Figure 8: Isobutyl radical variations

Regioselectivity refers to the preferential abstraction of hydrogen from specific carbon centers (*methyl, primary, secondary, or tertiary*) based on the relative stabilities of the radical character developed in the transition state.

For chlorination, the **early transition state exhibits minimal radical character**, meaning that the differential stabilization between primary and tertiary radicals is not yet fully manifested at the transition state geometry. Consequently, chlorination displays poor regioselectivity, abstracting hydrogen atoms from primary, secondary, and tertiary positions at comparable rates.

Bromination, in contrast, proceeds through a **late transition state with substantial radical character**, wherein hyperconjugative stabilization of the developing radical center significantly influences the E_a . Because tertiary radicals benefit from greater hyperconjugative stabilization than primary radicals, the transition state leading to the tertiary radical is preferentially stabilized, resulting in highly regioselective hydrogen abstraction from tertiary positions.

Experimental selectivity data support this mechanistic interpretation, as shown in Table 2.

C-H Bond	F (25°C, gas)	Cl (25°C, gas)	Br (150°C, gas)
CH ₃ - H	0.5	0.004	0.002
RCH ₂ - H	1	1	1
R ₂ CH - H	1.2	4	80
R ₃ C - H	1.4	5	1700

Table 2: Relative reactivities of C-H bonds toward halogen radicals (*normalized to primary C-H*)